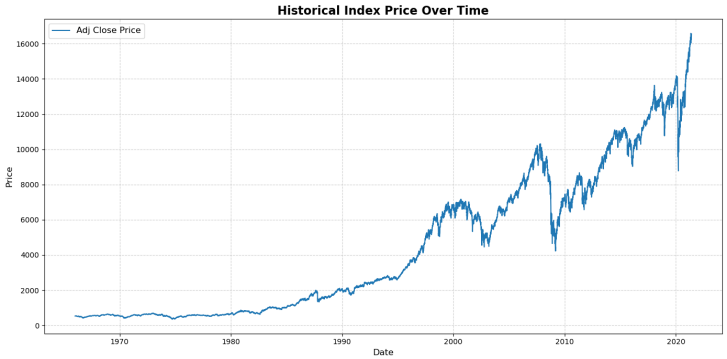
Visualize the Trends:

Plotting the the Close or Adj Close price over time to visually identify long-term trends and major market events.

Plotting Historical Prices

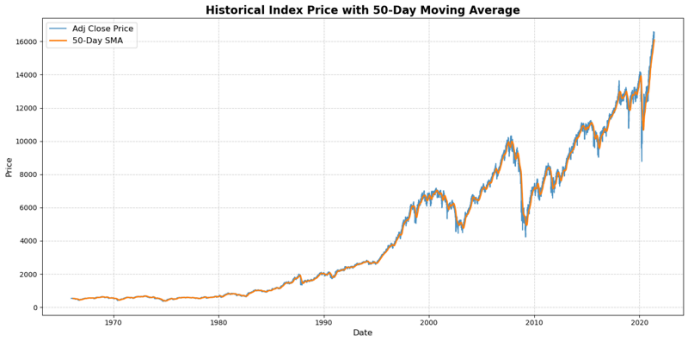


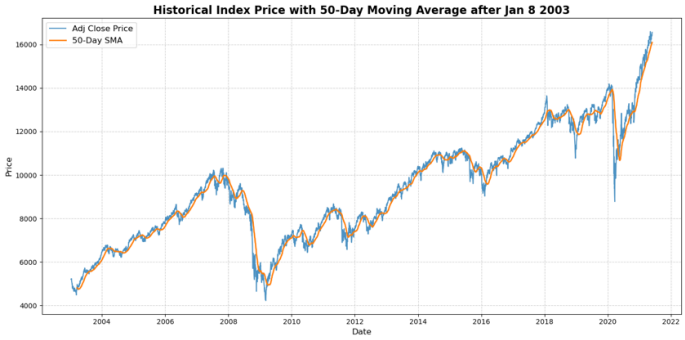
NYA Index Price with Moving Average Overlay.

The Lag: The orange SMA_50 line will smoothly trail behind the blue Adj Close line because it represents a 50-day historical average.

Missing Data at the Start: The SMA_50 line will not begin until the 50th day of your dataset, as it needs 50 data points to calculate the first average.

In NYA Index analysis, volume acts as a lie detector: a major price breakout on low volume is often a false signal, while a breakout on high volume indicates strong market conviction.





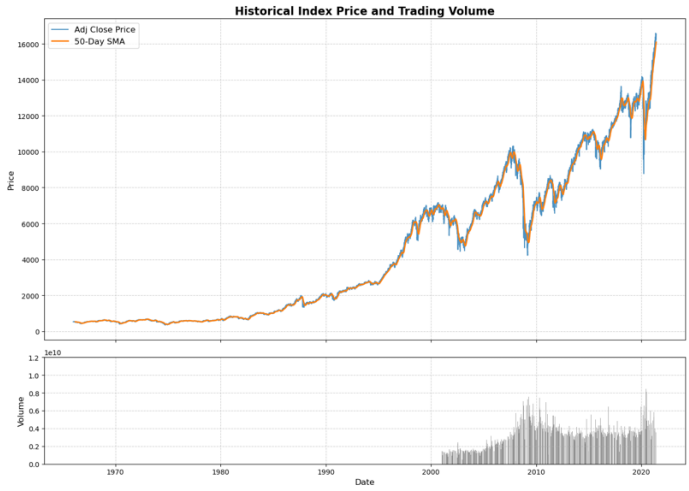
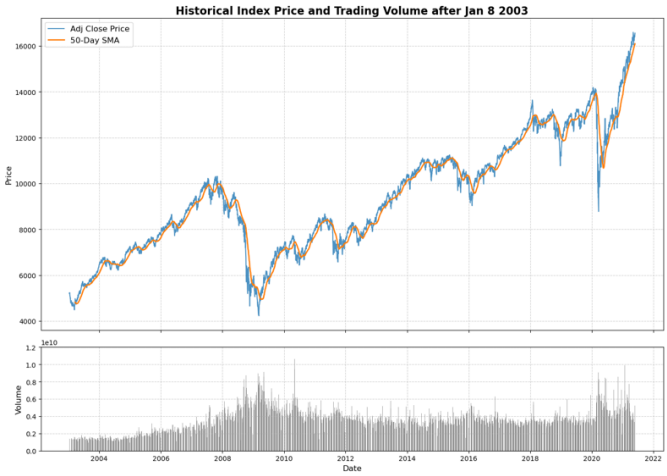
Daily Returns Distribution

Plotting the distribution of daily returns is a crucial step in financial data analysis. Traditional finance models often assume that returns are normally distributed (a perfect bell curve). However, real-world markets frequently experience "fat tails"—meaning extreme events (like market crashes or massive rallies) happen much more often than a normal distribution would predict.

What are we looking for?

The Peak: Financial data usually has a much sharper, higher peak around zero than a normal distribution. This means most days, the market barely moves.

The Tails: Look at the far left and far right edges of the chart. If the blue area (actual returns) extends further out or sits higher than the dashed red line (normal distribution), your index has "fat tails." This means extreme gains and extreme losses are more common than traditional statistics would expect.

The Skewness: Does the blue distribution lean more to the left or right? A longer left tail indicates that extreme negative returns (crashes) are larger/more frequent than extreme positive returns.

Price and Trading volume over time